

Standardised Mean Difference: Hedges' g

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Introduction

Hedges' g is the small-sample-corrected standardised mean difference (SMD), the preferred effect-size metric for meta-analysis of continuous outcomes. Cohen's d — the unstandardised SMD — over-estimates the population effect in small samples; Hedges' correction factor J de-biases it. The two are nearly identical for studies with combined sample size above 20 or so, but g should be used routinely whenever any included study is small.

Prerequisites

A working understanding of standardised effect sizes (Cohen's d), pooled standard deviation, and the bias of d in small samples.

Theory

Cohen's d is

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{\text{pool}}}, \quad s_{\text{pool}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}.$$

Hedges' correction is $J = 1 - 3/(4 \text{df} - 1)$ (with $\text{df} = n_1 + n_2 - 2$), and $g = Jd$. The correction is close to 1 for $\text{df} > 20$ but can be substantially below 1 for very small studies. The variance of g is

$$\text{Var}(g) = J^2 \left[\frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2)} \right].$$

Assumptions

Outcomes are approximately Normal within groups; group variances are approximately equal (use **SMDH** for unequal-variance variants); independent observations.

R Implementation

```
library(metafor)

set.seed(2026)
n <- 10 # small study
x1 <- rnorm(n, mean = 1, sd = 1)
x2 <- rnorm(n, mean = 0, sd = 1)

# Compute g and its variance via metafor
es <- escalc(measure = "SMD",
             m1i = mean(x1), sd1i = sd(x1), n1i = n,
             m2i = mean(x2), sd2i = sd(x2), n2i = n)
```

```
es
```

```
# Manual computation
s_pool <- sqrt(((n - 1) * var(x1) + (n - 1) * var(x2)) / (2 * n - 2))
d <- (mean(x1) - mean(x2)) / s_pool
J <- 1 - 3 / (4 * (2 * n - 2) - 1)
c(d = d, g = d * J)
```

Output & Results

`escalc(measure = "SMD")` returns Hedges' g (in the `yi` column) and its variance (`vi`) per study, ready for inverse-variance meta-analysis. The manual calculation confirms the J -correction; for $n_1 = n_2 = 10$, $J \approx 0.92$, reducing $d \approx 1.04$ to $g \approx 0.96$.

Interpretation

A reporting sentence: "Hedges' g for the simulated small study was 0.92 (95 % CI 0.01 to 1.83), corresponding to a large effect by Cohen's benchmarks ($d = 0.5$ medium, $d = 0.8$ large); the small-sample correction reduced the raw SMD from 1.04 to 0.92, reflecting the upward bias inherent in d for small studies." Always state the use of g vs d , especially when sample sizes are small.

Practical Tips

- Always use Hedges' g rather than Cohen's d for meta-analysis when any included study has $n_1 + n_2 < 20$; the bias correction is meaningful in that range.
- `metafor::escalc(measure = "SMD")` computes g and its variance in a single call; this is the standard workflow.
- For unequal variances (Levene's test rejects), use `measure = "SMDH"` which computes g using a pooled SD based on the separate group SDs.
- Report 95 % CIs alongside g ; the effect size alone does not convey precision, and meta-analysis weighting depends on the precision.
- For pre-post or within-subject designs, use SMCR or SMCC variants in `metafor::escalc()`; they account for the within-subject correlation that ordinary SMD ignores.
- For binary outcomes, switch to log-odds ratio or risk ratio; SMD is for continuous outcomes only.

R Packages Used

`metafor::escalc()` and `rma()` for the canonical effect-size and meta-analysis workflow; `meta::metacont()` for an alternative interface with continuous-outcome conventions; `metan` for a tidymodels-flavoured wrapper; `dmetar` for additional meta-analysis tools and visualisations.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics

- Network meta-analysis